

$$F(x) = -\frac{1}{4}x^4 + 3$$

$$-\pi \leq x \leq \pi$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} \left(-\frac{1}{4}x^4 + 3 \right) dx$$

$$= \frac{1}{\pi} \left(-\frac{1}{4} \int_{-\pi}^{\pi} x^4 dx + 3 \int_{-\pi}^{\pi} dx \right)$$

$$= \frac{1}{\pi} \left[-\frac{x^5}{20} + 3x \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[-\frac{\pi^5}{20} + 3\pi - \left(-\frac{\pi^5}{20} + 3\pi \right) \right]$$

$$= \frac{1}{\pi} \left(-\frac{\pi^5}{10} + 6\pi \right)$$

$$a_0/2 = 3 - \frac{\pi^4}{20}$$

$$a_0 = -\frac{\pi^4}{10} + 6 = 6 - \frac{\pi^4}{10}$$

usamos integración por partes al ser un producto.

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \left(-\frac{x^4}{4} + 3 \right) \cos(nx) dx$$

$$u = -\frac{x^4}{4} + 3$$

$$du = -x^3 dx$$

$$dv = \cos(nx) dx$$

$$v = \frac{\sin(nx)}{n}$$

$$\left(-\frac{x^4}{4} + 3\right) \left(\frac{\sin(nx)}{n}\right) - \int \frac{-x^3 \sin(nx)}{n} dx$$

$$= \left(-\frac{x^4}{4} + 3\right) \left(\frac{\sin(nx)}{n}\right) + \frac{1}{n} \int x^3 \sin(nx) dx$$

$$u = x^3 \quad dv = \sin(nx) dx$$

$$du = 3x^2 dx \quad v = \frac{\cos(nx)}{n}$$

$$\int x^3 \sin(nx) dx = \frac{x^3 \cos(nx)}{n} - \frac{3}{n} \int x^2 \cos(nx) dx$$

$$\int x^2 \cos(nx) dx = \frac{x^2 \sin(nx)}{n} - \frac{2}{n} \int x \sin(nx) dx$$

$$u = x^2 \quad dv = \cos(nx)$$

$$du = 2x dx \quad v = \frac{\sin(nx)}{n}$$

$$\int x \sin(nx) dx = -\frac{x \cos(nx)}{n} + \frac{1}{n} \int \cos(nx) dx$$

$$u = x \quad dv = \sin(nx) dx$$

$$du = dx \quad v = \frac{-\cos(nx)}{n}$$

$$= -\frac{x \cos(nx)}{n} + \frac{\sin(nx)}{n^2}$$

$$\int x \sin(nx) dx = -\frac{x \cos(nx)}{n} - \frac{x \cos(nx)}{n^2} + \frac{\sin(nx)}{n^3}$$

$$\int x^2 \cos(nx) dx = \frac{x^2 \sin(nx)}{n} + \frac{2x \cos(nx)}{n^2} + \frac{2x \cos(nx)}{n^3} - \frac{2 \sin(nx)}{n^4}$$

$$\int x^3 \sin(nx) dx = \frac{x^3 \cos(nx)}{n} - \frac{3x^2 \sin(nx)}{n^2} - \frac{6x \cos(nx)}{n^3} - \frac{6x \cos(nx)}{n^4} + \frac{6 \sin(nx)}{n^5}$$

$$\frac{1}{n} \int x^3 \sin(nx) dx = \frac{x^3 \cos(nx)}{n^2} - \frac{3x^2 \sin(nx)}{n^3} - \frac{6x \cos(nx)}{n^4} - \frac{6x \cos(nx)}{n^5} + \frac{6 \sin(nx)}{n^6}$$

$$a_n = \frac{1}{\pi} \left[\frac{\left(-\frac{1}{4}x^4 + 3\right) \sin(nx)}{n} - \frac{x^3 \cos(nx)}{n^2} + \frac{3x^2 \sin(nx)}{n^3} + \frac{6x \cos(nx)}{n^4} - \frac{6 \sin(nx)}{n^5} \right]_{-\pi}^{\pi}$$

$$a_n = \frac{1}{\pi} \left[\left(\frac{\left(-\frac{1}{4}\pi^4 + 3\right) \sin(n\pi)}{n} - \frac{\pi^3 \cos(n\pi)}{n^2} + \frac{3\pi^2 \sin(n\pi)}{n^3} + \frac{6\pi \cos(n\pi)}{n^4} - \frac{6 \sin(n\pi)}{n^5} \right) - \left(\frac{\left(-\frac{1}{4}(-\pi)^4 + 3\right) \sin(-n\pi)}{n} - \frac{(-\pi)^3 \cos(-n\pi)}{n^2} + \frac{3(-\pi)^2 \sin(-n\pi)}{n^3} + \frac{-6\pi \cos(-n\pi)}{n^4} - \frac{6 \sin(-n\pi)}{n^5} \right) \right]$$

$$a_n = \frac{1}{\pi} \left[-\frac{\pi^3 (-1)^n}{n^2} + \frac{6\pi (-1)^n}{n^4} - \frac{\pi^3 (-1)^n}{n^2} - \frac{6\pi (-1)^n}{n^4} \right]$$

$$a_n = \frac{1}{\pi} \left[-\frac{2\pi^3 (-1)^n}{n^2} + \frac{12\pi (-1)^n}{n^4} \right]$$

$$a_n = (-1)^n \left(-\frac{2\pi^2}{n^2} + \frac{12}{n^4} \right)$$

Como la función es par b_n es igual a 0

$$f(x) = 3 - \frac{\pi^4}{20} + \sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{2\pi^2}{n^2} - \frac{12}{n^4} \right) \cos(nx)$$